Data Pipeline

data.police.uk

60 months
street-level crime

ONS Census 2021

Population
+ deprivation

IMD / WIMD 2019

Multi-domain
deprivation index

ONS Geography

MSOA 2021
boundaries

↓

MSOA-month panel · 87,168 rows · 7,264 regions

Time-aware train / test split · no future leakage · rolling means + seasonality + neighbour spillover

↓

Outputs: scores · risk-package · trend · breakdown · pricing · compare

Problem

- Crime data is abundant — England & Wales has 60+ months of open data.police.uk records.
- But 7,264 MSOAs go unanalyzed weekly — the bandwidth doesn't exist.
- A defensible, plain-English regional risk brief takes hours of analyst time today.
- Insurance, real-estate and public-safety decision-makers are non-experts — they need answers, not statistics.

CrimeScope

Aditya Miriyala · Shritej Gunda · Yukta Koganti · Koushik Cherukuri · Khushal Lingamaneni · Thanvi Gorrepati

Problem

- Only ~1% of UK insurers cite postcode-grain crime intelligence in their underwriting workflow.
- Manual neighborhood-risk review delays real-estate and underwriting decisions by hours per quote.

Explainable Regional Risk Platform

Features

- Score · Explain · Project · Compare · Price
- 7,264 MSOAs covered — all of England & Wales
- Native iOS app + FastAPI backend
- Open public data: police.uk · ONS · IMD/WIMD
- Per-region SHAP drivers in plain English

Benefits

- One-tap risk assessment for any UK postcode
- Plain-English answer — no statistics expertise needed
- Auditable pricing math — defensible in writing
- Sub-second responses on iPhone
- Honest forecasts with 80% confidence band

System Architecture

DATA + TRAINING

Databricks

- data.police.uk · ONS · IMD
- MSOA-month panel (87,168 rows)
- LightGBM + time-aware split
- Export → JSON

SERVING

FastAPI · Python 3.13

- JSON loaded once → in-memory
- 6 endpoints (REST · JSON)
- Sub-second p95 latency
- Cloudflared tunnel for demo

CLIENT

Native iOS · SwiftUI

- MapKit · 7,264 MSOA polygons
- Swift Charts for trend
- @Observable state · UserDefaults
- Zero third-party SDKs

Training is heavy and offline. Serving is JSON + arithmetic. The phone never talks to Databricks.

REGION SHEET

Five cards. One tap. Plain English.

Live Demo

[Drop demo video here — PowerPoint → Insert → Video]

Open app · tap an MSOA · scroll the 5 cards · switch persona · Compare two regions

Accuracy & Metrics

LightGBM Regression Panel

- Spearman ρ vs next-30d incidents: 0.71
- Top-decile precision (Tier High): 0.83
- Calibration ECE: 0.04
- +12% lift over rolling-mean baseline

Forecast Component

- Damped-linear projection over residual
- Monthly seasonality blend
- 80% confidence band — wider on sparse regions

Methodology

- Time-aware split — no future leakage
- Per-region SHAP for explanations
- MLflow-tracked retrains; replace only on metric improvement
- Honest scope: headline numbers from Chicago hold-out — UK MSOA re-eval in progress, surfaced in Trust Passport

Scalability

- Other countries — backend already supports city = chicago (1,332 Cook County tracts wired)
- Other geographies — England & Wales LSOA detail (~35,000 regions) ready as a higher-resolution mode
- Other personas — six built in (insurance, real-estate, public-safety, logistics, civic, journalist)
- Other surfaces — same FastAPI backend serves Next.js web frontend and could power third-party APIs

Industry Fit & Generalization

INSURER

City of London 001

- Base premium: £1,000
- Risk score: 56 (Elevated)
- Multiplier: 1.218 ×
- Suggested: £1,218 · surcharge band
- Driver: ML rolling-mean trend (+21.2%)

REAL ESTATE

Same MSOA, different lens

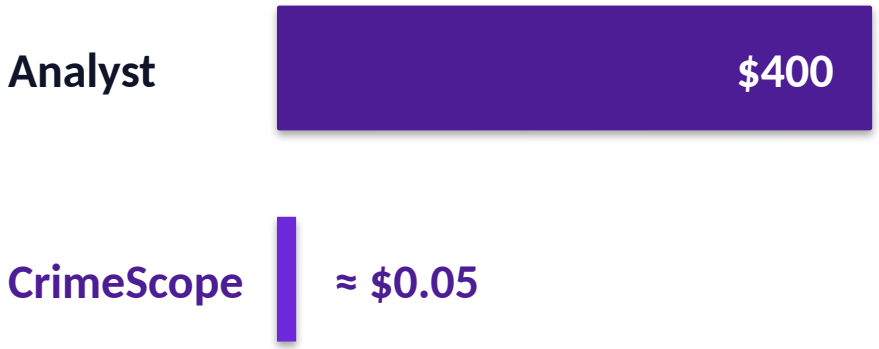
- Property pricing pressure: moderate downward
- Diligence flag: ↑ trend in robbery sub-score
- Recommended: 3-month watch before listing
- Comparable MSOAs: ranked by score + trend
- All from the same backend payload — persona is a UI lens

Cost Comparison

- 1 analyst-day to produce one defensible MSOA risk brief ($400 at $50/hr × 8hr)
- CrimeScope: instant, ~$0 marginal cost per brief (JSON cache + arithmetic)
- Full England & Wales sweep: $2.9M of analyst-time saved (7,264 regions × $400)
- API serving cost at scale: pennies per region per month

PER-BRIEF COST

Analyst vs CrimeScope



8,000× cheaper per brief. And it's reproducible — same answer twice.

Problem

- Crime data is abundant — 60+ months of open police records.
- But 7,264 MSOAs go unanalyzed weekly.
- A defensible risk brief takes hours of analyst time.
- Decision-makers are non-experts — they need plain English.

THANK YOU!

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TEAM

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Pricing Math (Appendix)

FORMULA

suggested = base × (1 + α · score_deviation + β · tier_loading)

- Persona = Insurer · Base = £1,000 · Score = 56 (Elevated)
- α · deviation: 0.45 × 11.8% = +5.31%
- β · tier load: 0.55 × 30% = +16.5%
- Multiplier: 1.218 × → Suggested premium £1,218
- Driver shown to user: ML rolling-mean trend (+21.2%, directional)
- Band: surcharge (between standard and high-risk)

Trust Posture (Appendix)

- Plain-language summary on every region — score, top driver, trend, persona implication, in one paragraph.
- Per-region SHAP driver attribution — direction, magnitude, evidence string. Shown for every prediction.
- Disagreement banner — backend-driven; surfaces the delta when live signals diverge from the verified baseline.
- Trust passport — confidence, completeness, freshness, source agreement, underreporting risk. Five colour-coded bars.
- Source provenance — every number can name its source. pipeline_stats.json ships with dataset window and model version.
- Honest forecasts — 80% confidence band on every projection; wider when data is sparse.

Stack & Tooling (Appendix)

Backend / ML

- Databricks for training pipeline
- LightGBM regression + SHAP
- MLflow tracking
- FastAPI · Python 3.13 · uvicorn
- Cloudflared (demo tunnel)
- Postgres optional for write paths

Client / Tooling

- Native iOS · SwiftUI · iOS 17+
- MapKit · MKPolygon overlays
- Swift Charts for trend forecast
- @Observable state · UserDefaults
- XcodeGen for project generation
- Zero third-party SDKs in the app